Project: Worked examples
Title: PCB trace width (IPC-2221)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the minimum width of a printed-circuit trace for a current and an allowed temperature rise by the legacy IPC-2221 curve-fit, conservative for multilayer boards. It also gives the resistance, voltage drop and I^2R loss of the trace as drawn.

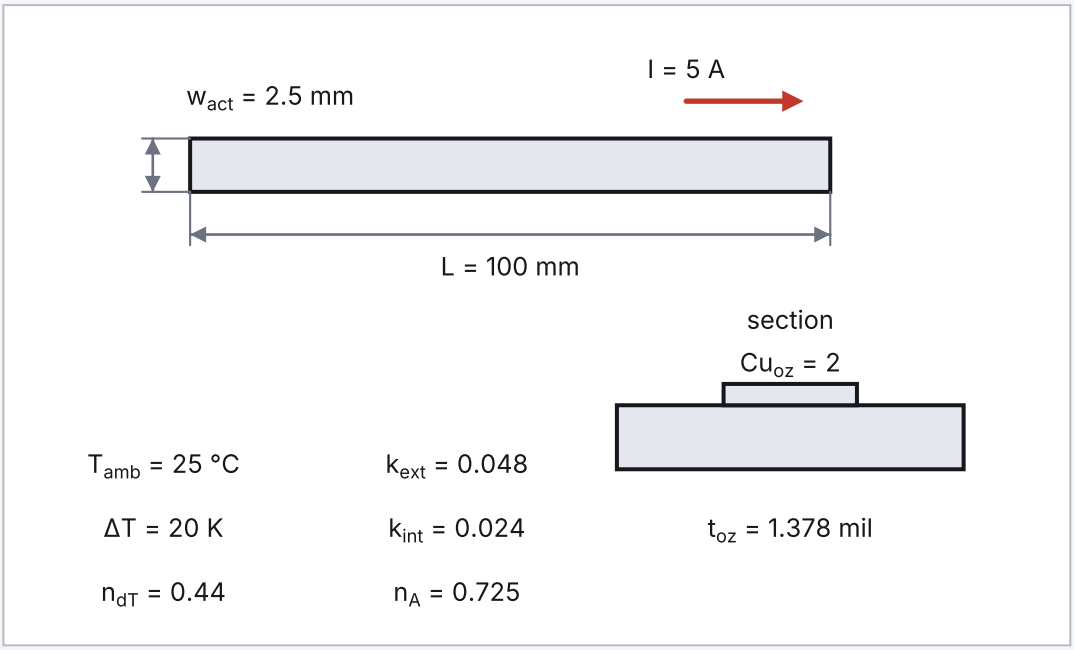


Figure 1 — A trace of width w_{act} and length L carrying I on an external layer: the copper on the board's face in section. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Trace cross-section	I = k·ΔT^0.44·A^0.725, so A = [I/(k·ΔT^0.44)]^(1/0.725) with A in mil², k = 0.048 external, 0.024 internal
Copper thickness	$t = w_{Cu} \cdot t_{oz}$, t_{oz} = 1.378 mil per oz/ft² (35 µm, derived)
Minimum width	$w_{min} = A/t$; utilisation $u_w = w_{min}/w_{act}$
Resistivity at the operating temperature	$\rho_T = \rho_{20} \cdot [1 + \alpha \cdot (T_{amb} + \Delta T - 20)]$
Trace resistance	$R = \rho_T \cdot L/(w_{act} \cdot t)$; voltage drop ΔV = I·R; loss $P = I^2 \cdot R$

Basis: Electrical properties: current-carrying capacity of printed-board conductors (legacy curve-fit; edition to be checked)[1](cl. 6); Resistivity and temperature coefficient of resistance[2](Ch. on DC circuits, 12th ed. (edition to be checked)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layer	Layer	External layer		drawing: PCB-LAY-014 rev B	any
w_{act}	Trace width as drawn	2.5	mm	drawing: PCB-LAY-014 rev B	0.05 … 10 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L	Trace length	100	mm	drawing: PCB-LAY-014 rev B	1 … 1000 mm
Cu_{oz}	Finished copper weight in oz/ft² (0.5 to 6), not a thickness in µm: 17.5 µm is ½ oz, 35 µm is 1 oz, 70 µm is 2 oz, 105 µm is 3 oz	2		spec: Fabrication note FN-7: 2 oz/ft² finished copper on the outer layers	0.5 … 6
ρ_{20}	Copper resistivity at 20 °C (annealed copper, 1.724e-8 Ω·m)	0.01724	Ω·mm²/m	reference: record: annealed copper 1.724e-8 Ω·m at 20 °C, derived physical constant	0.0165 … 0.0185 Ω·mm²/m
α	Temperature coefficient of resistance of copper	0.00393	1/K	reference: record: annealed copper, 0.00393 per K at 20 °C, derived physical constant	0.0035 … 0.0042 1/K
I	Trace current (the curve-fit was drawn for up to 35 A: above it the sheet is refused)	5	A	spec: Driver datasheet DD-12: 5 A continuous motor supply	0.01 … 35 A
T_{amb}	Ambient temperature at the board	25	°C	spec: Environmental specification ES-2: 25 °C at the board	-40 … 125 °C
dT	Allowed temperature rise above ambient	20	K	spec: Board design rule BDR-3: 20 K allowed rise on power rails	1 … 100 K
k_{ext}	Curve-fit constant k, external layer	0.048		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.01 … 0.1
k_{int}	Curve-fit constant k, internal layer	0.024		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.005 … 0.05
n_{dT}	Exponent of the temperature rise	0.44		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.3 … 0.6

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
n_A	Exponent of the cross-section	0.725		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.5 … 1
t_{oz}	Copper thickness per oz/ft² (35 µm = 1.378 mil, derived)	1.378	mil	reference: record: IPC-2221 legacy curve-fit constants; 1.378 mil = 35 µm per oz/ft², derived	1.2 … 1.5 mil

3. Assumptions

A1. The curve-fit is the legacy IPC-2221 relation for a trace with no plane or convective help. (IPC-2221 curve-fit (legacy); IPC-2152 is not implemented.)
A2. The curve-fit was drawn for 1 to 3 oz copper and currents up to 35 A. (IPC-2221 curve-fit (legacy).)
A3. The curve-fit was drawn for rises of 10 to 100 K and widths up to 10 mm. (IPC-2221 curve-fit (legacy).)
A4. The result screens the trace width and is not a thermal analysis of the board. (Scope of the curve-fit.)
A5. The curve-fit is conservative for multilayer boards. (IPC-2221 curve-fit (legacy).)
A6. Copper thickness is the nominal 1.378 mil (35 µm) per oz/ft² of finished copper. (Derived: 35 µm / 25.4 µm per mil.)
A7. Plating build-up on outer layers and etch-back are not modelled. (Scope of the sheet.)
A8. The resistance uses the trace as drawn at the temperature ambient plus rise. (Derived constants for annealed copper.)
A9. The resistivity at 20 °C is corrected with a linear temperature coefficient. (Derived constants for annealed copper.)
A10. Vias, pads, bends and skin effect are ignored. (Scope of the sheet.)
A11. The current is steady direct current, or RMS current at low frequency. (Scope of the curve-fit.)
A12. The rise is the allowed rise in the worst case. (Scope of the curve-fit.)

4. Working

[14] $k = k_{ext}$ [3((legacy))]
$= 0.048$
$= 0.048$
Curve-fit constant k (external layer)

[15] $I_n = I / (1\ A)$
$= 5\ A / 1\ A$
$= 5$
Current as a number of amperes

[16] $dT_n = dT / (1\ K)$
$= 20\ K / 1\ K$
$= 20$
Temperature rise as a number of kelvin

[17] $A_{req} = (I_n / (k * dT_n^{n_{dT}}))^{(1 / n_A)} * (1\ mil^2)$ [3((legacy))]
$= (5 / (0.048 \times 20^{0.44}))^{(1 / 0.725)} \times 1\ mil^2$
$= 98.511\ mil^2$
Minimum conductor cross-section

[18] $t_{Cu} = Cu_{oz} * t_{oz}$
$= 2 \times 1.378\ mil$
$= 70.002\ um$
Copper thickness

[19] $w_{min} = A_{req} / t_{Cu}$
= 98.511 mil² / 70.002 um
= 0.9079 mm
Minimum trace width
[20] $u_w = w_{min} / w_{act}$
= 0.9079 mm / 2.5 mm
= 0.36316
Width utilisation (minimum / as drawn)
[21] $A_{act} = w_{act} * t_{Cu}$
= 2.5 mm × 70.002 um
= 0.17501 mm²
Cross-section of the trace as drawn
[22] $T_{trace} = T_{amb} + dT$
= 25 degC + 20 K
= 45 degC
Trace operating temperature
[23] $\rho_T = \rho_{20} * (1 + \alpha * (T_{trace} - 20\ degC))$
= 0.01724 ohm·mm²/m × (1 + 0.00393 1/K × (45 degC - 20 degC))
= 0.018934 ohm·mm²/m
Copper resistivity at the operating temperature
[24] $trace_R = \rho_T * L / A_{act}$
= 0.018934 ohm·mm²/m × 100 mm / 0.17501 mm²
= 10.819 mohm
Trace resistance at the operating temperature
[25] $vdrop = I * trace_R$
= 5 A × 10.819 mohm
= 54.095 mV
Voltage drop along the trace
[26] $trace_{Ploss} = I^2 * trace_R$
= (5 A)^2 × 10.819 mohm
= 270.47 mW
Power loss in the trace (I^2R)

5. Results

SYMBOL	VALUE	DESCRIPTION
$trace_R$	10.82 mΩ	Trace resistance
$trace_{Ploss}$	270.5 mW	Power loss
vdrop	54.09 mV	Voltage drop
A_{req}	98.51 mil²	Minimum cross-section
w_{min}	0.9079 mm	Minimum trace width (governing)
u_w	0.36	Width utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Minimum trace width against the width as drawn	0.9079 mm	2.500 mm	0.36	[3]((legacy))

7. Applicability

WARNING records: tabulated values not taken from a reference record are k_{ext}, k_{int}, n_{dT}, n_A, t_{oz} (IPC-2221 legacy curve-fit method constants); ρ_{20}, α (Copper material constants). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

```
Template elec.pcb-trace@1.3.0 Content hash a52a92c6c8ccbf82 Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. IPC-2221B:2012.
2. Hughes E. Hughes Electrical and Electronic Technology. 12th ed. Pearson; 2016.
3. IPC-2221 cl. 6.